

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Strongest overall match. INJONGO is text-only **[Q43]** aligning with Telegram/SMS deployment, and dataset inspection confirms Ethiopic Unicode renders correctly without artifacts in all 19 sampled Amharic examples **[D29, D5]**. Annotation occurred on LabelStudio in native script **[Q43]**. Mixed Ethiopic/Arabic numeral usage is observed incidentally (D27), reflecting a pattern relevant to deployment without being formally designed. Two minor concerns moderate the score: (a) Unicode normalization form (NFC/NFD) is undocumented (gap_id 5 NOT FOUND), which could cause tokenizer mismatches in deployment, and (b) eng config exhibits transliteration degradation (D10: 'Dashen' → 'dasion'), which affects translate-test cross-lingual evaluation but not the primary in-language Amharic deployment path. SMS-specific signal degradation is not addressed in documentation, but per elicitation Telegram is the dominant primary channel and the deployment explicitly excludes romanized fallback (Q4 elicitation).

ID	Evidence
Q4	"We develop INJONGO—the first large-scale multicultural intent detection and slot-filling dataset covering 16 African languages, and English language." (p.1)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D10	raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኝላችሁ?" text: "can you please check my dasion bank account" — BANK_NAME: dasion bank — Original Amharic cites Dashen Bank; English translation renders "dasion" (spelling error) — shows transliteration degradation in eng config
D27	"ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ።" — DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car for 3 months starting next month." — Arabic numeral "3" mixed with Ethiopic in DATE span
D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)

Strengths

- Text-only modality matches Telegram/SMS deployment exactly **[Q43]**
- Ethiopic script renders correctly in all 19 sampled Amharic examples without encoding artifacts **[D29, D5]**
- Native-script annotation on LabelStudio **[Q43]**
- Incidental coverage of mixed Ethiopic/Arabic numerals (D27) reflects realistic input pattern
- License (CC BY 4.0 **[Q23]**) permits adaptation for production fine-tuning

ID	Evidence
Q23	"Dataset is released under CC BY 4.0 license." (p.2)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution match is good for primary channel: text in Ethiopic Unicode confirmed (D29). SMS-specific degradation (encoding loss, length truncation) is not represented in benchmark, but deployment scopes SMS as secondary and full Ethiopic as required [elicit^Q4].

IF-2: Yes — Telegram and Ethio Telecom infrastructure support Ethiopic Unicode reliably; 4G coverage exceeds 80% in primary deployment cities [WEB-6, WEB-8]; Telegram transmits Ethiopic reliably (YAML).

IF-3: Numeral mixing observed (D27); Unicode normalization undocumented (gap_id 5); SMS Ethiopic transmission reliability may vary by carrier — not benchmarked. eng-config translate-test path shows transliteration noise (D10) that would affect cross-lingual transfer evaluation but not in-language deployment.

IF-4: Minor form gaps: (a) NFC/NFD normalization unspecified — could cause tokenizer subword mismatch; (b) no SMS-degraded variants tested; (c) romanized Amharic explicitly excluded from deployment scope so absence is acceptable.

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WEB-3	McGill-NLP/Injongo GitHub — Amharic example አባይ ባንክ confirms Ethiopic rendering at release (https://github.com/McGill-NLP/Injongo)
WEB-6	4G coverage >80% in primary deployment cities — text channel infrastructure available (https://www.mordorintelligence.com/industry-reports/ethiopia-telecom-mno-market)
WEB-8	Ethio Telecom FY2024/25 — 1,683 new mobile sites, 512 cities with 4G (https://addisinsight.net/2025/07/24/ethio-telecom-reports-162-billion-birr-revenue-and-4-9-trillion-birr-in-telebirr-transfers-in-2017-fiscal-year/)

Information Gaps

- Unicode normalization form (NFC/NFD) of release files (gap_id 5 NOT FOUND)
- Whether SMS channel Ethiopic delivery reliability has been measured by EEU/AAWSA
- Byte-offset vs character-offset behavior for span annotations in Ethiopic UTF-8 (dataset analysis Limitation 5)

Requires Expert Verification

- Whether tokenizers used in deployment match the byte-level normalization of INJONGO release files
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Remediation

Gap 1: Unicode normalization form (NFC vs NFD) of INJONGO Amharic release is undocumented, which can cause tokenizer subword mismatch in production.

Recommendation: Audit the parquet files for Unicode normalization, normalize to a single form (NFC recommended for Ethiopic), and verify tokenizer behavior on Telegram-sourced production input matches the normalized training data.